PAGASA SYNOP Validator

Ruleset v0.16.1-review

Rainfall popup and saved-record cross-checks. This addendum takes precedence over the attached v0.16.0 review rules for the changes described here.

Simple rainfall review

Review rainfall / saved records opens a separate popup. At 00 UTC, it displays the four six-hour periods for the preceding UTC day. At other hours it displays the current UTC day so far; future periods cannot be edited. Records are separated by station and explicit UTC date. Save and close stores completed-period amounts locally; validate the observation afterward.

Occurrence evidence

Manually confirmed rainfall and usable coded rainfall amounts are saved as supporting evidence with their accumulation period. Intermediate evidence wholly inside the next main report period checks for a contradictory zero or absent rainfall group. At 00 UTC, evidence within the preceding 24 hours cross-checks the scheduled 6RRR4 and Section 3 7RRRR groups. Missing groups produce a review warning; a zero amount conflicting with saved rain produces an error. These checks never select the manual rainfall checkbox or require present weather solely because rain occurred earlier in the day.

Amounts and incomplete records

Only exact six-hour amounts are added to form the 24-hour total. Intermediate amounts are not added again. Missing is distinct from zero and Trace; incomplete periods cannot fully verify a total. A mixed trace and measurable-rain day requires a confirmed measured total. Historical evidence does not establish the timing of rainfall inside an interval that extends outside the target period.

Retention and limits

Pressure history remains eligible for 36 hours; rainfall remains available for 8 days in this browser. This update does not add monthly accumulation, export/import, cloud sharing, or a permanent archive. Existing pressure, cloud, visibility, CB, gust, duplicate and group-order rules are retained.

Review ruleset v0.16.0-review

PAGASA SYNOP Validator | 11 October 2026

This review release extends the GitHub v0.15.0 baseline. The following addendum takes precedence where behavior changes. The complete v0.15.0 ruleset follows for all unchanged meteorological checks.

Pressure basis and required groups

Tayabas (98427) and Tanay (98433) use station pressure from Section 1 3P0P0P0P0 for both the 3-hour 5appp check and the 24-hour 58/59 check, following the practices supplied by the station reviewer. Other stations retain MSLP as their comparison basis.

Station pressure and MSLP are decoded and saved separately. Comparisons require the same station, UTC observation time and pressure basis. A missing previous value produces an unavailable-comparison warning, not a numerical mismatch. Differences must agree exactly to 0.1 hPa.

A usable station-pressure group is required at main observation times and at every observation for the two configured stations. Format and duplicate checks remain active. Values outside 500-1100 hPa are errors. Broad elevation-aware review ranges warn outside 850-1085 hPa for Tayabas and 800-1020 hPa for Tanay. These are review settings, not instrument calibration limits.

Pressure decoding below 1000 hPa

The thousands digit is omitted in the pressure group. For example, 39917 is 991.7 hPa. This release corrects a baseline decoding error for pressure values below 1000 hPa.

Highlights inside the editable code

After validation, errors have red backgrounds and warnings have amber backgrounds behind the original editable text. Clicking or keyboard-activating a finding selects the first affected token. Missing groups have no text to mark and remain explained in the findings. Editing the report clears old highlights until it is validated again. Copy returns plain original text.

Exact rainfall and history

At 00, 06, 12 and 18 UTC the observer can enter the exact previous-six-hour rainfall to 0.1 mm, including measured zero or Trace. The amount does not select the previous-six-hour weather checkbox. That checkbox remains manual and resets only when a different observation is entered.

At 00 UTC the review lists four nonoverlapping six-hour periods ending at 06, 12, 18 and 00 UTC. The periods are matched by station and full UTC date. The observer can review and edit entries before saving. Missing periods cannot be treated as zero.

When every period has a numeric measured amount, the page sums integer tenths and compares the result with the Section 1 6RRR4 and Section 3 7R24R24R24R24 groups. A separately entered measured 24-hour total must agree with a complete numeric sum. A measured total can provide a comparison when historical periods are missing, but it does not make those periods complete.

Trace remains distinct from zero. A day with only zero and trace entries is a trace day. Mixed trace and measurable rain requires a measured 24-hour total because trace has no exact numeric amount. Section 5 2RRRR at 00 UTC is checked against the exact six-hour amount. At 06/12/18 UTC the scheduled Section 1 6RRR1 group is checked instead.

RRR coding: 990 is trace, 991-999 are 0.1-0.9 mm and numeric amounts of 1 mm or more use whole millimetres. The 24-hour 7-group retains tenths of a millimetre. Saturation codes are handled separately from literal amounts.

Local retention and date checks

Pressure readings remain eligible for 36 hours after saving. Exact rainfall remains eligible for 8 days. Both use this browser profile on this site and do not sync between computers. An explicit UTC observation date must match the day in AAXX, which has no month or year. Confirm the suggested date before saving.

Closing the normal browser or restarting normally preserves saved local history. Clearing site data, using private browsing, storage restrictions or changing browser/device/site domain can remove it or make it unavailable. The saved history is not a backup of the complete SYNOP text.

Scope and sources

Monthly rainfall accumulation is deferred. Existing monthly group format and reporting-time checks remain in the baseline. The page cannot verify observations that were not entered or measurements that the observer has not confirmed.

GitHub baseline: rein217/pagasa-synoptic-validator, main commit b90a4ca1541e44aa34846ba3838fba1e98424a28. Station identifiers: PAGASA WIS2 station collection at wis.pagasa.dost.gov.ph/oapi/collections/stations/items. Browser storage: developer.mozilla.org/en-US/docs/Web/API/Window/localStorage. Pressure-group definition: WMO Manual on Codes, FM 12, 12.2.4.

PAGASA SYNOP Validator

Operational Checking Ruleset

Version 0.15.0 - Reviewed Branch Update

Document status	Operational review guide
Coding form	WMO FM 12 SYNOP (AAXX), with PAGASA practices
Validator input	One complete observation at a time
Published build	Ruleset v0.15.0
Basis	v0.14.3 operational rules plus reviewed collaborator improvements

Purpose

This document describes the checks implemented in the PAGASA SYNOP Validator for operational review, regression testing, and controlled maintenance. It does not replace the official WMO Manual on Codes, PAGASA observing instructions, or observer judgment.

Important: 'No coded errors detected' means only that the observation passed the rules currently implemented. It is not a certification that every observation or coding decision is correct.

1. Inputs, result levels, and bulletin schedule

Input	Purpose
Complete SYNOP code	Required AAXX observation, preferably with bulletin heading and '=' end marker.
MSLP 3 hours earlier	Exact sign-category and ppp-magnitude check for 5appp.
MSLP 24 hours earlier	Exact signed check of Section 3 58/59 at 00 and 12 UTC.
Rainfall occurred	Manual indicator for rainfall during the six hours immediately before observation.

Finding levels

• **Error:** conflict with required syntax, schedule, arithmetic, or an implemented operational rule.

• **Warning:** possible code that needs observational review or missing supporting information.

• **Information:** an optional section or check is unavailable.

• **OK:** no implemented error or warning was produced.

Bulletin schedule

• SMPH20 is expected at 00, 06, 12, and 18 UTC.

• SIPH20 is expected at 03, 09, 15, and 21 UTC.

• When present, bulletin YYGG is compared with AAXX YYGGiw.

2. Structure, syntax, and duplicate groups

Element	Implemented check
AAXX / YYGGiw / IIiii	Required positions; valid day, hour, iw, and five-digit station identifier.
Mandatory groups	iRixhVV and Nddff immediately follow the station identifier.
Sections	333 begins Section 3 and 555 begins Section 5.
Group format	Five figures or permitted solidi; missing '=' produces a warning.
Section 3 order	Standard Section 3 groups are checked for indicator order.
Section 1 order	After iRixhVV and Nddff, recognized groups must follow indicator order (1, 2, 3, and so on).

Semantic duplicate detection

Single-occurrence Section 1 families are compared by their meaning, not only by exact text. Thus **10313 10312** is an error because both are 1snTTT air-temperature groups. The same approach applies to dew point/humidity, station pressure, MSLP, pressure tendency, present/past weather, and the main cloud group. Legitimately repeatable groups remain allowed.

An Nddff-shaped group found after the mandatory Nddff position is also reported. Example: **70701 23602** is flagged because 23602 contains a valid wind direction 36 but is placed as another wind group.

3. Pressure and local history

Rule	Check
4PPPP plausibility	Configured operational range: 870.0 to 1085.0 hPa.
5appp magnitude	Absolute current-minus-3-hour difference equals ppp exactly to 0.1 hPa.
5appp sign	Positive: a=0-3; negative: a=5-8; zero: a=0,4,5.
24-hour change	At 00/12 UTC, 58 is positive/zero and 59 is negative; exact 0.1 hPa agreement.
Zero change	59000 is rejected; zero uses 58000.

Browser-local MSLP history

- A single structurally valid and plausible 4PPPP value is retained for 36 hours.
- Matching uses station identifier and SYNOP day/hour; manual entries take priority.
- Month-boundary matching is skipped when month/year cannot be known from YYGGiw.
- History stays only in the current browser/device and is not synchronized.

If previous pressure is unavailable, the validator requests it with a warning and does not create a false mismatch.

4. Rainfall, weather, and visibility

Topic	Implemented rule
iR and location	Checks required presence and placement of 6RRRtR in Sections 1 and/or 3.
tR schedule	tR=4 at 00; tR=1 at 06/12/18; tR=7 at 03/09/15/21 UTC.
Manual choice	The code does not auto-select the checkbox. A 24-hour total cannot locate rain within the latest six hours.
Reset	A different YYGGi/station clears the box; corrections within the same observation preserve it.
RRR	001-989 are whole millimetres; 990 is trace; 991-999 are 0.1-0.9 mm. Thus 60074 is 7 mm, tR=4.
Weather	Cross-checks rainfall against applicable ww and W1/W2 indications.

Visibility rules

Phenomenon	Horizontal visibility treatment
Light precipitation	Suggested 9 km or more; lower value gives a warning.
Moderate precipitation	2 km to less than 9 km.
Heavy precipitation	2 km or less; a value above 2 km is an error.
Present weather 40	2 km or less; a value above 2 km is an error.
Present weather 04-06	No mandatory upper limit. Haze-intensity ranges are guidance only.

5. Cloud rules and reportable layers

Cloud checks combine Nddff, h in iRixhVV, 8NhCLCMCH, and Section 3 8NsChshs groups.

PAGASA 1-3-5 selection

• Report the lowest layer at 1 okta or more.

• Report the next higher layer at 3 oktas or more.

• Report the next higher layer at 5 oktas or more.

• Always report Cumulonimbus when not already selected; maximum four reportable groups.

• Order reportable groups from lower to higher bases.

Independent layer amounts

Each Ns is estimated as if other layers did not exist, so layers can overlap. The validator does not add Section 3 Ns values to derive Nh. It does check that an individual low-cloud Ns does not exceed Nh. CB does not imply that no other low cloud exists.

Example: **84908 333 81915 83820 87280 94945** is consistent with Nh=4, one okta CB and three oktas Cumulus.

Sky state

N	Treatment
0	Omit main and individual cloud groups.
9	Omit main cloud group; report 89/hshs vertical visibility.
/	Omit main and individual cloud groups.

6. CB, gustiness, and monthly rainfall

949CD nature codes

C	Meaning
4	Isolated Cumulonimbus
5	Numerous Cumulonimbus
6	Isolated Cumulus and Cumulonimbus
7	Numerous Cumulus and Cumulonimbus

D is direction 0-9. Multiple 949CD groups may be used for CB in different directions. A valid example is **94968**: isolated Cumulus and Cumulonimbus to the north.

Gustiness

A 904tt or 907tt group must be followed immediately by 911ff and 915dd. Orphan speed or direction groups are reported.

Monthly rainfall

At 00 UTC on day 1, Section 5 may contain the six-hour rainfall group and the previous-month total. When the monthly value exceeds 1000 mm, the extended form is **6//// RRRRR**. The five-digit total is interpreted in tenths of a millimetre.

7. Release safeguards and limitations

Regression coverage

• Exact 3-hour and 24-hour pressure differences, including 0.1 hPa failures.

• Main/intermediate bulletin mismatches and bulletin/AAXX time consistency.

• Section 1 out-of-order groups and additional Nddff candidates.

• Heavy precipitation and ww=40 at and above the 2 km boundary.

• Valid and invalid high-cloud obscuration cases.

• Combined Cu/CB 949 codes, unrestricted ww=04-06 visibility, and semantic duplicate groups.

• Manual rainfall reset, same-observation preservation, and RRR trace/decimal decoding.

Known limitations

• The validator sees coded text, not the original instrument readings or sky observation.

• Local pressure history cannot infer month/year and is not shared between devices.

• Suggestions require observer review; the program does not rewrite or transmit an observation automatically.

Maintenance workflow

• Edit routine values in dist/js/ruleset-config.js.

• Edit meteorological logic in dist/js/ruleset.js and interface behavior in dist/js/app.js.

• Add a correct and incorrect regression example for every rule change.

• Run: node tests/ruleset-regression.test.mjs

• Review the v0.15.0 branch before merging it into main.

Operational status

This is an operational review build. Continue comparison with official WMO and PAGASA documentation and record every accepted correction in the changelog.